

OroTan-MX | Competitive Analysis Matrix

India Market Edition — Expanded Comparator Set (v2)

Table 1 — Three Core Clinical Dimensions

Product	Anaesthesia	Astringency (stabilized, high-load)	Barrier Protection (extended mucoadhesion)	Gap / Note
Orajel Benzocaine 20% — US/export	✓	✗	✗	Pure anaesthetic only; washes away rapidly; no astringency or mucoadhesion
Mucopain Benzocaine 20% I.P. — ICPA	✓	✗	✗	Indian analogue of Orajel; instant numbing (10–20 min); no barrier, no tannin matrix
Dologel CT Choline Sal 8.7% + Lignocaine 2% + BAC — Dr. Reddy's	✓	✗	~	Salicylate class only; no tannin cross-linking or phytobioenhancer
Zytee / Zytee-L Choline Sal 9% + BAC ± Lignocaine — Raptakos Brett	✓	✗	~	Same profile as Dologel CT; salicylate class only; no tannin cross-linking or phytobioenhancer
Hexigel Chlorhexidine Gluconate 1% — ICPA	✗	✗	~	Antiseptic / antimicrobial; no anaesthetic, no tannin astringency; some mucosal contact time
Kank-A Benzocaine + film-former — US/export	✓	~	✓	Astringent load present but not stabilized / not high-tannin concentration
Bonjela Choline Sal + Cetalkonium Cl — Reckitt, export	✗	✓	✗	Astringency via generic salicylates, not tannin cross-linking; weak bioadhesion; no rapid anaesthetic
Gengigel Hyaluronic Acid — export	✗	✗	✓	Hyaluronan-based barrier only; no anaesthetic, no tannin matrix
Generic Tannic Acid Gels Unbranded	✗	✓	✗	Astringent but unstable — phase-separates; poor shelf-life; no pharmacopeial anaesthetic
OroTan-MX Lignocaine HCl 2% + Tannin matrix + Piperine-Gingerol	✓	✓	✓	3-in-1 — stable high-tannin mucoadhesive with pharmacopeial anaesthetic + bioenhancer adjunct

Table 2 — Formulation & Technical Differentiators

Feature	Orajel / Mucopain (Benzocaine class)	Dologel CT / Zytee (Choline Sal class)	Hexigel (CHX class)	Market-Standard Gels (Typical)	OroTan-MX
Stable high-tannin load (no polymer complexation / phase separation)	✗	✗	✗	✗	✓
Sting-free / low-osmolality formulation	~	~	~	✗	✓

Feature	Orajel / Mucopain (Benzocaine class)	Dologel CT / Zytee (Choline Sal class)	Hexigel (CHX class)	Market-Standard Gels (Typical)	OroTan-MX
Standardized, HPLC/GC-MS verified actives	✗	✗	✗	✗	✓
Permeation-enhancing bioenhancer adjunct (piperine-gingerol)	✗	✗	✗	✗	✓
In-vitro mucoadhesion testing vs. porcine buccal mucosa	✗	✗	✗	Not reported	✓
ICH-aligned stability data package	✗	✗	✗	Not reported	✓*
Dual anaesthetic mechanism (pharmacoepial grade lidocaine + astringent tannin)	✗	~	✗	✗	✓
India regulatory origin / scheduled pharma compliance	✗	✓	✓	✗	✓

✓ = Present / Confirmed	✗ = Absent / Gap	~ = Partial / Claimed but not fully substantiated	✓* = Proposed milestone — not yet completed data (see Notes)
-------------------------	------------------	---	--

- Notes & Caveats**
- ICH-aligned stability (✓*)**: Described in source as 'ICH-aligned stability potential' — a proposed validation milestone, not completed data. Asterisked to signal this to regulatory reviewers.
 - Mucopain / Dologel CT / Zytee / Hexigel** are India-specific products added in this v2 update. Their gaps are inferred from published compositions and are consistent with the general 'market-standard gels' categorization in the original source document.
 - Formulation basis**: This matrix describes the Lignocaine HCl 2.0% w/w formulation (Version 2 / Business Development Proposal). The open question between Version 1 (eugenol) and Version 2 (lignocaine) applies equally here.
 - Dologel CT / Zytee**: Choline salicylate provides NSAID anti-inflammatory action, not tannin-mediated astringency. BAC (benzalkonium chloride) contributes short-contact antisepsis, not extended mucoadhesion.
 - Hexigel**: Chlorhexidine gluconate 1% confers antimicrobial / antiseptic action with some mucosal contact time, but does not provide pharmacoepial-grade anaesthesia or tannin astringency.